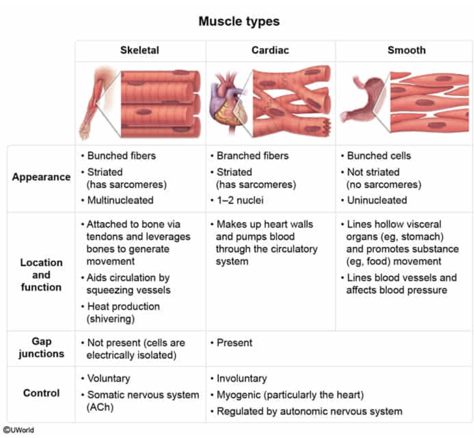
Human cardiac muscle differs from skeletal muscle because it:

- A. appears striated. [8%]
- B. requires acetylcholine to contract. [3%]
- ✔ C. contains intercalated discs. [73%]
- D. contains cells with multiple nuclei. [13%]

Correct
74% answered correctly

**Explanation:**




The walls of the heart are composed primarily of **cardiac muscle fibers**, which function to generate powerful, coordinated contractions that are responsible for the continuous pumping of blood through the vessels of the circulatory system. Microscopically, cardiac muscle appears **striated** due to the regular arrangement of actin and myosin filaments into small contractile units called sarcomeres, and each cardiac muscle cell contains one or two nuclei. Cardiac muscle is unique in that each cell is connected to adjacent cells via **intercalated discs**, which are regions of cell contact that contain both desmosomes (to prevent cells from separating during contraction) and gap junctions (to facilitate direct ion exchange for synchronized contraction).

**(Choice A)** Cardiac and skeletal muscle both appear striated due to the presence of sarcomeres.

**(Choice B)** Cardiac muscle is under involuntary control and is myogenic, meaning that it *does not require* nervous system input to contract. Instead, specialized cells in the sinoatrial node spontaneously depolarize to produce electrical impulses that spread through the cardiac muscle via its intercalated discs. However, cardiac muscle contraction can be *regulated* by neural and hormonal input. For example, parasympathetic signaling slows the rate of contraction, and sympathetic and hormonal signaling can increase heart rate. Skeletal muscle, on the other hand, is under voluntary control and must be stimulated by the acetylcholine-releasing motor neurons of the somatic nervous system to contract.

**(Choice D)** Cardiac and skeletal muscles are *alike* in that both contain cells that have multiple nuclei. Cardiac muscle cells typically have no more than 2 nuclei, but skeletal muscle cells, which fuse during development, can contain hundreds of nuclei.

**Educational objective:**
Muscle tissue can be classified as cardiac, skeletal, or smooth. Cardiac muscle is striated and under involuntary control. Skeletal muscle is striated and under voluntary control, and smooth muscle is not striated and under involuntary control.

**Time spent:** 0 sec
**Subject:** Biology

**QID:** 401449
**System:** Musculoskeletal System

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